19 July 1972
SUPERSEDING
MIL-C-12668C
30 June 1967

MILITARY SPECIFICATION

CANNON, 120MM GUN: M58

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a type of cannon intended for installation in a heavy tank.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARD

Military

MIL-STD-109 - Quality Assurance Terms and Definitions

DRAWING

U. S. Army Weapons Command

F7248892 - Indicator Gage

PUBLICATIONS

U. S. Army Weapons Command

EPL 8766509 - Engineering Parts List for Cannon, 120MM Gun: M58
DL 8766509 - List of Inspection for Cannon, 120MM Gun: M58
LSQAP 8766509 - Master List of Supplementary Quality Assurance Provisions for Cannon 120MM Gun: M58
PDS 8766508 - Packaging Data Sheet, Tube for Cannon 120MM Gun: M58

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. Unless otherwise specified (see 6.1.1), the first article shall be included in the pilot pack (see 5.1).

3.2 Materials, design and construction. The cannons and parts thereof shall conform to this specification, to MIL-C-13931, and to the drawings and specifications listed in Engineering Parts List, EPL 8766509.

3.3 Shrinkage. Except for a distance of 16 inches rearward from the shoulder on the tube where the first shrinkage contact is desired, the surface of the tube to be enveloped by the jacket shall first be coated with copper sulphate solution (blue vitriol) and then given a thin uniform coat of lubricant. The lubricant shall consist of powdered graphite and light machine oil mixed to the consistency of paint.

3.4 Breech mechanism. The breech mechanism shall operate without evidence of interference or erratic movement whether operated manually or semi-automatically. Camming of the operating crank during counter-recoil or manipulation of the operating handle to open the breech shall rotate the breechblock crank rearward and downward, dropping the block to open the breech. The extractors shall operate to eject the empty cartridge case and shall latch the breechblock in the open position. The breech closing mechanism shall return the breechblock to the closed position when the extractors are tripped by insertion of a round or when the breech is closed manually.

3.4.1 Firing mechanism group. The cocking, percussion and trigger mechanisms shall function without binding, evidence of interference or erratic movement. During descent of the breechblock, the cocking-lever arm, forced by a camming surface on the breech ring, shall engage the retractor which in turn shall force the guide and firing-pin rearward, compressing the firing-spring and allowing the sear to latch the guide and firing-pin in the cocked position. Rearward movement of the trigger plunger shall pivot the trigger, allowing the sear to rotate and release the percussion mechanism. The compressed firing-spring shall force the firing pin to protrude beyond the face of the breechblock and thereby compress the retracting spring which shall then withdraw the firing-pin behind the front face of the breechblock.

3.4.2 Firing pin guide. Overtravel of the firing pin guide when measured as specified in 4.5.1.4 shall be not less than 0.06 inch.

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3.4.3 Firing pin protrusion. The firing pin protrusion, measured from the front face of the breechblock, shall be not less than 0.11 inch nor more than 0.16 inch.

3.5 Performance.

3.5.1 High pressure resistance. The cannon shall be capable of withstanding pressure of 113% ± 4% of the 57600 psi upper pressure limit at 70°F when tested as specified in 4.5.2.2

3.5.2 Functioning. There shall be no impairment in the functioning of the cannon components during and after testing specified in 4.5.2.1.

3.5.3 Material soundness. All parts shall be free from cracks and fractures and shall comply with any soundness requirements on the applicable drawing.

3.5.4 Bore. Bore diameters, measured across lands shall not exceed the basic (minimum) diameter specified on the applicable drawing by more than the amount specified in Table I.

Table I Bore enlargement

Distance forward of commencement of rifling	0.10 in.	1 in.	1 cal.	2 cals.	3 cals. to muzzle
Amount (inch)	0.009	0.009	0.008	0.007	0.006

NOTE: Interpolate to determine the bore enlargement between distances given.

3.6 Workmanship. Workmanship shall be as required in MIL-C-13931.

QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions are as specified in MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

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4.4 First article inspection. A first article shall be submitted for inspection in accordance with the contract requirements. The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspections specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Examinations.

4.5.1.1 Inspection provisions. Cannons shall be inspected for conformance to requirements of the provisions of this specification, MIL-C-13931, SQAP's listed in Master LSQAP 8766509, and SQAP-APPENDIX-WVT. Cannon in the quantities indicated shall be subjected to the tests described herein. Any failure to conform to requirements or acceptance criteria shall be cause for rejection.

4.5.1.2 Shrinkage. The assembly of the jacket and tube during the shrinkage process shall be under the surveillance of the Government representative to the extent necessary to assure compliance with 3.3 and the applicable drawings.

4.5.1.3 Firing pin protrusion. Firing pin protrusion shall be measured on each cannon with an approved gage. Cannons failing to meet the requirements of 3.4.3 shall be cause for rejection.

4.5.1.4 Firing pin guide overtravel. Overtravel of the firing pin guide shall be measured on each cannon with Gage 7248892 or an approved equivalent. The procedure shall be as follows: Remove the firing spring retainer, insert the gage and lock it in place. Lower the breechblock slowly to the point where the sear engages the firing pin guide, as indicated by an audible click. At this point adjust and set the dial indicator of the gage to zero. Continue lowering the breechblock until the maximum reading on the gage is obtained. This reading is the measurement of the overtravel. Cannons failing to meet the requirements of 3.4.2 shall be cause for rejection.

4.5.1.5 Workmanship. Workmanship shall be examined in accordance with MIL-C-13931.

4.5.1.6 Final examination. Final examination of each cannon shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract.

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4.5.1.7 Inspection of preparation for delivery.

4.5.1.7.1 Examination. The classification of defects and the AQLs for examination of the packaging, packing, and marking are as follows:

CATEGORY	DEFECT	AQL	INSPECTION METHOD
Major			
101	Illegible or incorrect marking	1.0	Visual
102	Improper level of preservation, packaging or packing	1.0	Visual
103	Inadequate cleaning or drying	1.5	Visual
104	Improper preservative application	1.5	Visual
105	Missing or improper cushioning	1.5	Visual
106	Improper blocking or bracing	1.5	Visual/Manual
107	Improper closure or strapping of container	1.5	Visual/Manual
Minor			
201	Workmanship	4.0	Visual

4.5.1.8 Inspection approval stamp. The application of the inspection approval stamp shall be examined as specified in MIL-C-13931.

4.5.2 Tests.

4.5.2.1 Functioning tests. Each cannon shall be operated manually and semi-automatically to test smoothness of operation. Cocking action, trigger actuation, firing mechanism action, and breech mechanism operation tests shall be performed to ascertain proper functioning. Extraction shall be tested with an empty service cartridge case and with the cannon in a horizontal position. Cannons failing to meet the requirements shall be cause for rejection.

4.5.2.2 Performance test. Each cannon shall be subject to performance testing to assure compliance with 3.5, by a Government activity in accordance with procedures specified by the engineering and quality assurance elements of the procuring activity.

4.5.2.2.1 Firing tests. When specified in the contract, cannon shall be subjected to proof firing tests by Government personnel at a proving ground designated by USATECOM. The first cannon or repair tube manufactured under a new contract shall be subject to the firing schedule shown on Table II. Subsequent cannon or repair tubes shall be proof fired with one round of 113 $\pm$ 4 percent upper pressure limit and to such production tests as prescribed by the responsible commodity arsenal to ascertain conformance with specification and performance requirements.

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TABLE II

Number of Rounds	Upper Pressure Limit, %
2	100
2	113 ± 4
2	100
2	113 ± 4

4.5.2.3 Functioning. After performance testing, each cannon shall be tested for proper functioning in accordance with 4.5.2.1. Cannon failing to meet the requirements of 3.5.2 shall be cause for rejection.

4.5.2.4 Material Soundness. After testing, each cannon and each repair tube shall be examined visually for freedom from cracks, fractures and other defects. Furthermore, breechblocks, breech rings and outer surfaces of tubes shall be magnetic particle inspected in accordance with applicable part drawings and specifications. Inner surface of tubes shall be examined by means of a borescope. Any evidence of cracks or fractures shall be cause for rejection.

4.5.2.5 Bore enlargement. After firing no more than 10 rounds, the bore diameters across the lands of each cannon and each repair tube shall be measured with an air gage, star gage or other approved gage. Measurements shall be recorded. Enlargement exceeding the limits specified in Table I shall be cause for rejection.

4.5.2.6 Records. Cannon and repair tubes that pass firing tests shall have applicable data entered on DA Form 2408-4, "Weapon Record Data" card by the proving ground. Cannon and repair tubes accepted on a proof sampling plan shall have applicable data recorded on DA Form 2408-4 by the contractor and approved by the Government representative.

4.5.3 Inspection equipment. Data lists containing equipment used in the inspection of parts, subassemblies, and assemblies are listed in DL 8766509.

4.5.3.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.1.1), responsibility for acquisition, maintenance, and disposition of acceptance inspection and test equipment prescribed in the List of Inspection Equipment DL 8766509, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.3.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

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5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract, a pilot pack (see 6.4) shall be forwarded in accordance with 3.1. Pilot packs for the tube shall be packaged to the level of packaging specified in the contract and packed level C in accordance with the requirements of Packaging Data Sheet 8766508.

5.2 Preservation, packing, and marking. The tube shall be unit packaged, packed, and marked in accordance with the requirements of Packaging Data Sheet 8766508, for the level of protection specified in the contract (see 6.1.1).

5.3 Repair parts. Repair parts shall be unit packaged, packed, and marked in accordance with the requirements of Packaging Data Sheet 8766508.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number and date of this specification.
b. Requirements for submission of the shipping instructions for first article (see 3.1 and 4.4) and pilot pack (see 3.1 and 5.1).

(1) The contract should designate the examinations and test to be performed by the contractor and the examinations and tests to be performed by the Government.

(2) The contract should specify the sample size (i.e., the number of units) of the first article.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(4) The contract should designate at whose expense a retest may be performed.

(5) The contract should designate that the acceptable first article shall be delivered in accordance with the terms of the contract, or

(6) Any additional or extended examinations and tests beyond the scope of this specification should be specified and identified as a separate item of the contract.

c. List of repair parts required.
d. Serial numbers to be used for the cannons and the repair tubes.
e. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.3.1).
f. Selection of applicable levels of preservation, packaging and packing (see 5.2).

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6.1.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

6.2 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate on the Management Control Systems Summary List, DD Form 1660.

6.3 Unless otherwise specified (4.5.3.1), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

6.4 Definition of pilot pack. A requirement in conjunction with a specification or contract for a first article in order to base approval of packages.

Custodian:
Army--WC

Preparing activity:
Army--WC

User activity:
Navy--MC

Project No. 1015-0242

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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